

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
5	(a)		Energy Conversion C0 - E_p to E_k C1 - and E_k to E_p C2 - Continued conversion, back and forth C3 - Decrease or increase linked to position e.g. A to B, E_k increases Energy Loss L0 - E_p or E_k degraded (or equivalent) L1 – Max height on right below A [or doesn't reach C] L2 - Final E_p and $E_k = 0....$ L3 -Linked to position i.e at B L4 - Energy lost as heat or internal energy L5 - Friction or air resistance linked to energy loss L6 - Molecular explanation of friction or air resistance 5-6 marks 7 – 11 of C and L marks are present <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks 4 – 6 of C and L marks are present <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks 1 – 3 of C and L marks are present <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	2	4		6		

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	(b)	(i)	Force in direction of travel required / $280 \cos 35^\circ$ needed Accept: force and distance not in the same direction Accept: horizontal component needs to be calculated or used Don't accept not all the 280 N is used in pulling the sled or pulled at an angle	1			1		
		(ii)	$280 \cos 35^\circ$ or 229.4 N (1) Substitution: $W = 229.4$ (ecf) $\times 3\,000$ (1) [$W = 6.88 \times 10^5$ J] Substitution: $P = \frac{6.88 \times 10^5}{1200}$ ecf on W and t (1) $P = 573.4$ W unit mark or suitable unit alternative (1) Alternative: $280 \cos 35^\circ$ or 229.4 N (1) Calculation of $v = 2.5 \text{ m s}^{-1}$ (1) Substitution: $P = Fv$ i.e. $P = 229.4$ (ecf) $\times 2.5$ (1) $P = 573.4$ W unit mark or suitable unit alternative (1)	1	1 1		4	4	
			Question 5 total	4	7	0	11	4	0